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United States Patent	12409781
Kind Code	B1
Date of Patent	September 09, 2025
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Rearview mirror with integrated handicap sign

Abstract

A rearview mirror with integrated sign (the “device”) being attachable to a rearview mirror of a vehicle. The device has at least one clamp, each with a moveable jaw and a fixed jaw, for attaching the device to the rearview mirror. The moveable jaw can be manually moved away from the fixed jaw to place the device on, or remove the device from, the rearview mirror. Upon placing the device in position on the rearview mirror, a spring basis the moveable clamp towards the fixed clamp with sufficient force to hold the device onto the rearview mirror. The device includes a sign with indicia thereon, for example a handicap symbol. The device includes a power drive assembly and/or manual drive assembly to selectively deploy the sign so that indicia on the sign is viewable or retract the sign to a stowed position that is principally within the housing.

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Appl. No.: 18/123806

Filed: March 20, 2023

Related U.S. Application Data

us-provisional-application US 63321662 20220318

Publication Classification

Int. Cl.: G09F21/04 (20060101); B60R1/12 (20060101); G09F7/20 (20060101); G09F7/18 (20060101)

U.S. Cl.:

CPC **B60R1/12** (20130101); **G09F7/20** (20130101); **G09F21/04** (20130101); G09F2007/1843 (20130101); G09F2007/1865 (20130101); G09F2007/1873 (20130101)

Field of Classification Search

CPC: B60R (1/12); G09F (7/20); G09F (21/04); G09F (2007/1843); G09F (2007/1865); G09F (2007/1873); G09F (11/18); G09F (11/29); G09F (15/0025); G09F (15/0062); G09F (17/00)

USPC: 297/97.1- 9; 40/514-517; 160/127

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/321,662, filed Mar. 18, 2022, which is incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates generally to handicap signs. More particularly, the invention is a rearview mirror for a vehicle that includes an integrated handicap sign. In an advantageous embodiment, the invention allows for a handicap sign to be selectively extended from a rearview mirror to a displayed position and retracted to a stowed position within the rearview mirror.

BACKGROUND OF THE INVENTION AND RELATED ART

(2) Handicap parking permits are designed to help individuals with conditions that impact their ability to walk short distances. The type of disability can range widely, from vision loss to a cardiac condition to an ambulatory limitation. One of the most common reasons people utilize handicap parking permits is because of limited mobility. This means that an individual is unable to walk short distances without the use of an assistive device like a cane, walker, or wheelchair.

(3) There are many conditions that can cause limited mobility. For example, chronic illnesses like lupus, inflammatory bowel disease, obesity and asthma can all result in an inability to walk far

distances. Many people who suffer arthritis, whether it's osteoarthritis, rheumatoid, or gouty, experience loss of mobility. Loss of mobility can also be due to a serious injury; particularly an injury to the spinal cord, which results in tremors or muscle weakness and even paralysis-all of which can impact the ability to walk or move. Additionally, neurodegenerative disorders such as Parkinson's or multiple sclerosis impair motor function which can greatly affect the ability to walk.

(4) Handicap signs are used to identify handicapped accessible and designated parking spots. Whereas handicap signs on a vehicle signify that an occupant of the vehicle is disabled and, therefore, the vehicle is allowed to be parked in a handicapped designated parking space. Typically, such a vehicle would have a rear license plate with a symbol thereon of a person in a wheelchair.

(5) In addition to a disability license plate, a handicap sticker or tag is often placed in or on the vehicle. This secondary notice allows for others to view the handicap status from a direction other than from behind the vehicle. For example, a handicap sticker may be placed on a windscreen or on the exterior body or fender of a vehicle.

(6) For various reasons, a person may desire to selectively either display or not display a handicap sticker. Disadvantageously, a sticker is intended to be permanent or at least remain affixed in position for an extended time. As such, a person cannot selectively and repeatedly move the sticker between an affixed display location and being removed, that is, non-displayed. Another disadvantage is that a sticker affixed to a windscreen blocks the line-of-sight therethrough and leaves a sticky residue when removed.

(7) Similar disadvantages occur by affixing the sticker to the body or fender of a vehicle. That is, the handicap sticker cannot be selectively and repeatedly affixed in a displayed position then removed from sight. And removal of the sticker would leave an undesired sticky residue and possibly damage the paint of the vehicle.

(8) Placards having a handicap symbol allow for the sign to be selectively displayed or stowed, thereby overcoming the permanency issue of stickers. Placards can be loose or in hang tag configuration with a hook for hanging from a rearview mirror. Nonetheless, placards have their own disadvantages.

(9) Loose placards are typically laid on top of the dashboard of a vehicle. However, by not being affixed to the vehicle, a loose placard can easily slide off the dash during driving and end up on the floor where it may be difficult for a person with a disability to return the sign to the dashboard. In this manner loose placards can become damaged or, since they are easily moved, lost.

(10) Hang tag placards are an improvement in that they can be selectively placed in a display position hanging from a rearview mirror or stowed. Nonetheless, the act of attaching to and removing the placard from the rearview mirror is inconvenient and requires a degree of dexterity that some disabled people may find difficult or impossible. Additionally, when hanging from the rearview mirror in the displayed position the placard partially blocks a driver's line of sight through the front windscreen which is a detriment to driving the vehicle. Additionally, when the placard is not being displayed it can easily be lost or become damaged.

(11) As outlined above, conventional handicap stickers and placards have disadvantages in not being, or in the manner that they are, selectively interchangeable between an in-use display position and a stowed position. Accordingly, in view of the aforementioned conventional handicap stickers and placards, there exists a need for an improved handicap sign that can be selectively displayed and stowed. There further exists a need for an improved handicap sign that remains attached to a vehicle regardless of whether the sign is being displayed or stowed. A still further need exists for a handicap sign that can be easily moved from the stowed position to the displayed position and vice versa.

SUMMARY OF THE INVENTION

(12) The invention is a rearview mirror with integrated sign (the "device") being attachable to a rearview mirror of a vehicle. The device has at least one clamp for attaching the device to the rearview mirror. Each of the at least one clamp has a moveable jaw and a fixed jaw. The moveable

jaw can be manually moved away from the fixed jaw to place the device on, or remove the device from, the rearview mirror. Upon placing the device in position on the rearview mirror, a spring basis the moveable clamp towards the fixed clamp with sufficient force to hold the device onto the rearview mirror. The device includes a sign with indicia thereon, for example a handicap symbol. The sign can be selectively deployed so that indicia on the sign is external of the housing of the device and viewable or retracted so that the sign is being stowed principally within the housing.

(13) In an aspect, the invention is embodied by a rearview mirror with integrated sign, having a housing; a mirror attached to the housing; at least one clamp attached to the housing, each of said at least one clamp having a moveable jaw and a fixed jaw being adapted for attachment to a rearview mirror; a holder assembly attached to the housing, wherein the holder assembly includes a sign shaft, and wherein the sign shaft is capable of rotational movement; a sign attached to the sign shaft, wherein the sign includes indicia; a slot in the housing adapted for allowing the sign to move there-through; and a power drive assembly functionally attached to the sign shaft and capable of rotating the sign shaft in a first direction to roll up the sign on the sign shaft in a stowed position and rotating the sign shaft in a second direction to deploy the sign so that indicia on the sign are external of the housing. In an embodiment, the power drive assembly includes a motor for rotating the sign shaft and a switch for selectively turning the motor on or off, and for selectively rotating the sign shaft in the first direction or the second direction. In a further embodiment, the device further includes at least one sensor, wherein the sensor signals to stop the motor when the sign reaches the deployed position when deploying the sign, and signals to stop the motor when the sign reaches the stowed position when retracting the sign.

(14) In an aspect, the invention is embodied by a rearview mirror with integrated sign, having a housing; a mirror attached to the housing; at least one clamp attached to the housing, each of said at least one clamp having a moveable jaw and a fixed jaw being adapted for attachment to a rearview mirror; a holder assembly attached to the housing, wherein the holder assembly includes a sign shaft, and wherein the sign shaft is capable of rotational movement; a sign attached to the sign shaft, wherein the sign includes indicia; a slot in the housing adapted for allowing the sign to move there-through; and a manual drive assembly functionally attached to the sign shaft and capable of rotating the sign shaft in a first direction to roll up the sign on the sign shaft in a stowed position and rotating the sign shaft in a second direction to deploy the sign so that indicia on the sign are external of the housing.

(15) In an aspect, the invention is embodied by a rearview mirror with integrated sign, having a housing; a mirror attached to the housing; a holder assembly attached to the housing, wherein the holder assembly includes a sign shaft, and wherein the sign shaft is capable of rotational movement; a sign attached to the sign shaft, wherein the sign includes indicia; a slot in the housing adapted for allowing the sign to move there-through; and a power drive assembly functionally attached to the sign shaft and capable of rotating the sign shaft in a first direction to roll up the sign on the sign shaft in a stowed position and rotating the sign shaft in a second direction to deploy the sign so that indicia on the sign are external of the housing.

(16) Other aspects, objects, features and advantages of the invention will be made apparent or will be readily understood and appreciated by those skilled in the related art as the invention is described in greater detail hereinafter and is shown in the accompanying drawing figures. It is envisioned that all such aspects, objects, features and advantages of the invention as shown and described herein will be within the intended broad scope of the appended claims. The above and other aspects, objects, features and advantages may be accomplished by any of the exemplary embodiments of the invention described herein and illustrated in the accompanying drawings. However, it should be appreciated that the drawing figures are for illustrative purposes only, and that many modifications, changes, revisions and substitutions may be made to the exemplary embodiments illustrated herein without departing from the broadest reasonable interpretation of the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The aforementioned objects, features and advantages of the invention will be more fully understood and appreciated when considered in conjunction with the accompanying drawings.
- (2) FIG. 1 is an environmental perspective view illustrating a rearview mirror with integrated handicap sign (the “device”) according to an exemplary embodiment of the invention.
- (3) FIG. 2 is an exploded bottom-rear-left perspective view of the rearview mirror with integrated handicap sign of FIG. 1.
- (4) FIG. 3A is a rear view of a back cover of the rearview mirror with integrated handicap sign of FIG. 1, showing a pair of clamps in a closed position for securing the device to a rear-view mirror of a vehicle according to an exemplary embodiment of the invention.
- (5) FIG. 3B is an interior front view of the back cover of FIG. 3A.
- (6) FIG. 3C is a rear view of the back cover of the rearview mirror with integrated handicap sign of FIG. 1, showing the pair of clamps in an open position for attaching or removing the device to a rear-view mirror of a vehicle according to an exemplary embodiment of the invention.
- (7) FIG. 3D is an interior front view of the back cover of FIG. 4C.
- (8) FIG. 4 is a fragmented top-rear perspective view of the rearview mirror with integrated handicap sign of FIG. 1, showing internal components in the right portion of the device according to an exemplary embodiment of the invention.
- (9) FIG. 5A is a bottom-rear perspective view of a holder assembly of the rearview mirror with integrated handicap sign of FIG. 1, for carrying a sign according to an exemplary embodiment of the invention.
- (10) FIG. 5B is an exploded bottom-rear perspective view of the holder assembly of FIG. 5A.
- (11) FIG. 6 is a fragmented top-rear perspective view of the rearview mirror with integrated handicap sign of FIG. 1, showing internal components in the left portion of the device according to an exemplary embodiment of the invention.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

- (12) The present invention will now be described more fully hereinafter with reference to the accompanying drawings in which preferred embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be considered as limited to the embodiments set forth herein. These exemplary embodiments are provided so that this disclosure will be both thorough and complete, and will fully convey the scope of the invention to those skilled in the art. Throughout the several views of the drawings, like reference characters designate the same or similar parts.
- (13) FIGS. 1-2 show a rearview mirror with integrated handicap sign (also referred to herein as the “device”), designated generally by reference character **10**, according to an exemplary embodiment of the invention. The device **10** allows for a sign **12** to be selectively extended therefrom to a display position and retracted to a stowed position within the device **10**. Although the sign **12** described and depicted herein is a handicap sign, it is to be understood that the device **10** can be used to display any type of sign **12**, slogan, artwork or the like.
- (14) The device **10** includes housing **14** having a front **18**; left and right sides **20**, **22**; a top **24**; a back cover **26**; and a base **28**. The front **18** includes a mirror **29** allowing for a driver of a vehicle to view rearwardly behind the vehicle. Preferably, the back cover **26** and/or base **28** can be detached from the main body **16** to allow access to repair internal components as needed.
- (15) In a preferred embodiment, the device **10** is an after-market product that is attachable via a pair of clamps **30** to a rearview mirror **32** of a vehicle. In an embodiment, each clamp **30** has a fixed jaw **34** and a slidably moveable jaw **36**. The clamps **30** are considered to be attached to the housing. Each fixed jaw **34** extends rearwardly from the back cover **26** a distance to span at least a

portion of the depth of the rearview mirror **32**, preferably about $\frac{1}{3}$ inch to $\frac{3}{4}$ inch, and has a downward directed lip **38** to hold the device **10** in place.

(16) Referring to FIGS. **2** and **3A-D**, each movable jaw **36** includes an elongated first section **40**, a second section **42** that extends perpendicular to and rearward from the first section **40** (and thusly rearward from the back cover **26**) a distance to span at least a portion of the depth of the rearview mirror **32**, preferably about $\frac{1}{3}$ inch to $\frac{3}{4}$ inch, and an upward directed lip **44** to hold the device **10** in place.

(17) The first section **42** of each moveable jaw **36** is slidably moveable within a respective channel **46** of the back cover **26**. In a preferred embodiment, the back cover **26** includes a raised portion **48** with a pair of openings **50** which form respective entries to the channels **46** through which respective second sections **42** are received. The raised portion **48**, lateral walls **52** and tabs **54** define the pair of channels **46** and limits movement of the second sections **42** such that each movable jaw **36** is moveable only towards or away from its respective fixed jaw **34**. By manually sliding the movable jaws **36** away from their respective fixed jaws **34** the distance therebetween is increased whereby the clamp **30** may be selectively attached to or removed from the rearview mirror **22**.

(18) A spring **56** has an intermediate section that is positioned by a knob **58** and opposed end section **56a**, **56b** that are respectively engaged with a respective hook **60** provided on the moveable jaws **36**. The spring **56** biases the moveable jaws **36** towards the fixed jaws **34** with sufficient force such that the clamp **30** holds the device **10** on the rearview mirror **32**. The knob **58** is considered to be part of the housing **14**.

(19) It is within the scope of the invention that the device **10** can alternatively be configured such that all the jaws may be constructed to be moveable in order to attach and remove the device to a rearview mirror **32**. For example, a similar arrangement described with the moveable jaws **36** may also be used instead of the fixed jaws **34**.

(20) It is also within the scope of the invention that the device **10** can alternatively be configured to be directly attached to the front windscreen **62** of a vehicle. The alternate embodiment the device **10** includes a stem and mounting base (for example, being similar to the stem **64** and mounting base **65** of the rearview mirror **32** that is shown in FIG. **1**) which attaches the device **10** to the windscreen **62**, instead of having clamps **30** for attaching the device **10** to a rearview mirror **32**.

(21) The device **10** includes a holder assembly **66** (FIGS. **5A**, **5B**) for carrying the sign **12** and a drive assembly having at least one of a power drive assembly **68** (FIG. **6**) for electronically deploying and retracting the sign **12** and a manual drive assembly **70** (FIG. **4**) for manually deploying and retracting the sign **12**.

(22) Referring to FIGS. **2**, **4**, **5A** and **5B**, the holder assembly **66** includes a collar **72** mounted in the housing **14** interior to the front **18**, a shaft **74** rotatably disposed within the collar **72**, opposed bearings **76**, **78** positioned on the end portions **80**, **82** of the shaft **74**, and a bevel gear **84** affixed to the terminal end of end portion **80** for engagement with the manual drive assembly **70**.

(23) The sign **12** is attached at its proximal end **86** to the shaft **74** by an adhesive or any other suitable fashion. A paper weight **88** is attached to the distal end **90** of the sign **12**, by an adhesive or any other suitable fashion, to provide sufficient weight so that the sign **12** hangs external of the housing **14** downwardly and generally flat when deployed, as shown in FIG. **2**. The housing base **28** has a slot **91** through which an intermediate portion of the sign **12** passes while deploying and retracting the sign **12**. The sign **12** is flexible, made of textile, paper, plastic, fabric or the like, so that it rolls up about the shaft **74** when the sign **12** is being retracted and is generally flat when the sign **12** is deployed. Indicia, words, images and the like (collectively and individually the "indicia **93**") are presented on the sign **12**. For example, a handicap symbol **93** may be printed on the front face and/or back face of the sign **12** so that the symbol **93** is viewable when the sign **12** is in the deployed position.

(24) Referring to FIGS. **2**, **4** and **6**, a pair of bearing housing **92**, **94** are attached to the base **28**.

Each bearing housing **92, 94** includes a race **96, 98** in which the respective bearing **76** or **78** resides. Each bearing housing **92, 94** includes a base portion **99, 100** having part of the race **96** or **98** and a cover portion **101, 102** having the remaining part of the race **96** or **98**. The cover portion **100, 102** is attached to the base portion **96, 98** by screws (not shown) which allows access to place the bearings **76, 78** into the respective race **96, 98** during assembly. (In FIG. 4, cover portion **101** is shown sectioned at line 4-4 with a portion of the cover portion **101** exploded from the base portion **99** and the remaining portion of the cover portion **101** in position behind the bearing **76** in order to show the positioning the cover portion **101** in position behind the bearing **76**).

(25) Referring to FIGS. 2 and 4, the manual drive assembly **70** can be used to deploy and retract the sign **12**. The manual drive assembly **70** includes a scroll wheel **104** rotatably attached to the back cover **26** of the housing **14** by a shoulder screw **106**. The scroll wheel **104** has a bevel gear **108** which engages and transmits rotational motion to bevel gear **84**, and an undulated peripheral edge **109** for being engaged by a person's finger when manually turning the scroll wheel **104**. The scroll wheel **104** is sized and positioned so that a portion of the scroll wheel **104** extends through a slot **110** in the base **28** for accessibility by a person. As such, a person may use their finger to selectively rotate the scroll wheel **104** counterclockwise which imparts rotational motion to the shaft **74**, via the bevel gears **84, 108**, causing shaft **74** to roll up the sign **12** into a stowed position principally within the housing **14**. Similarly, a person may use their finger to selectively rotate the scroll wheel **104** clockwise which imparts rotational motion to the shaft **74**, via the bevel gears **84, 108**, causing shaft **74** to unroll the sign **12** to the deployed position displaying indicia **93** on the sign **12**.

(26) Referring to FIGS. 2 and 6, the sign **12** can also be deployed and retracted using the power drive assembly **68**. The power drive assembly **68** includes a DC motor **112** that is affixed within the interior of the housing **14**, a motor shaft **114**, and a coupler **116** which joins the motor shaft **114** to the end portion **82** of the sign shaft **74** so that rotational motion from the motor shaft **114** is transmitted to the sign shaft **74** in order to deploy and retract the sign **12**. It is within the scope of the invention that other means may be used to join the motor shaft **114** to the sign shaft **74** such as, for example, gears or the like.

(27) Referring to FIG. 2, a charger (not shown) is plugged into a power outlet, for example a vehicle's cigarette lighter (not shown), to provide power to the device **10**. The charger includes a DC connector on each end for respective connection to the cigarette lighter at one end and at the other end to a DC jack **118** provided on the device **10**. Optionally, the charger can include an on/off switch to control whether current is provided to the device **10** while the device **10** is plugged into the cigarette lighter. In an alternative embodiment, the device may be powered by one or more batteries (not shown).

(28) Electronics include a switch **120**, an H-bridge **122**, a printed circuit board, a micro controller **124** and a voltage regulator **125** for turning the DC motor **112** on and off and controlling whether the DC motor **112** turns the sign shaft **74** clockwise to deploy the sign **12** or counterclockwise to retract the sign **12**. An exemplary switch is a DPDT rocker switch **120**, which has three states of ON, OFF and ON but the ON states are on opposite polarity. That is, the DPDT switch **120** can be used to make the motor shaft **114** rotate clockwise or counterclockwise by change the polarity of the voltage applied. An exemplary micro controller **124** is an Adurino Pro Mini.

(29) A pair of sensors **126, 128** are provided to signal the motor **112** to shut off when the sign **12** has reached a fully deployed position and when the sign has reached a fully retracted position. In an embodiment, the sensors are inferred sensors **126, 128**, each having a channel **130, 132** through which respective lateral edges of the sign **12** pass when the sign **12** is being deployed or retracted. An example of a suitable sensor is an optical endstop switch such as a TSCT **2103** switch. The sign **12** is provided with an opening **134** near its upper left corner. Upon activation, the motor **112** will continue to deploy the sign **12** until the sensor **126** detects that the opening **134** is positioned within the channel **130** and signals to the motor **112** to stop.

(30) Similarly, the sign **12** is also provided with an opening **136** near its bottom right corner. When retracting, the motor **112** will continue to retract the sign **12** until the opening **136** is positioned within the channel **132** and signals to the motor **112** to stop.

(31) It is to be understood that not all electrical components and wiring are illustrated or described herein as such is within the ordinary skill in the art.

(32) In operation, the device **10** is attached to a vehicle's rearview mirror **32** by manually sliding the moveable jaws **36** away from the fixed jaws **34** so that the span of the clamps **30** are greater than the height of the rearview mirror **32**. The clamps **30** are then placed about the top and bottom of the rearview mirror **32** and the moveable jaws **36** released. The moveable jaws **36**, urged by the spring **56**, are drawn towards the fixed jaws **34** with sufficient force to attach and hold the device **10** to the rearview mirror **32**. If desired, the device **10** can be removed from the rearview mirror **32** by manually sliding the moveable jaws **36** away from the fixed jaws **34** to remove the clamping force from the rearview mirror **32**.

(33) To provide power, one end of a charger is plugged in the vehicle's cigarette lighter, and the other end is plugged into the device's DC jack **118**. To deploy the sign **12**, a person presses the rocker switch **120** into the down position causing the motor **112** to turn the shaft **74** clockwise unwinding the sign **12**. The sign **12** continues to unwind until sensor **126** detects opening **134** in the upper left corner of the sign **12** whereupon the motor **112** is signaled to stop. In the deployed position, indicia **93** on the sign **12** is external of the device housing **14** and thus viewable to the public.

(34) When driving the vehicle, the sign **12** is placed in the retracted stowed position being principally within the device housing **14** in order to obviate potential obstruction of vision. To retract the sign **12**, a person presses the rocker switch **120** into the up position causing the motor **112** to turn the shaft **74** counterclockwise and winding the sign **12** about the shaft **74**. The sign **12** continues to be retracted until the inferred sensor **128** detects opening **136** in the lower right corner of the sign **12** wherein the motor **112** is signaled to stop.

(35) Alternatively, a person can deploy and retract the sign **12** by manually using the scroll wheel **104**. Accordingly, a person can manually rotate the scroll wheel **104** to cause the shaft **74** to turn clockwise thereby unwinding the sign **12** to the deployed position. And a person can manually rotate the scroll wheel **104** to cause the shaft to turn counterclockwise thereby winding up the sign **12** in the stowed position. Thusly, a person could still use the device **10** even if they do not have a charger. Although it is preferred that the device **10** include both the power drive assembly **68** and a manual drive assembly **70**, the device **10** may be configured with only one of them.

(36) Regardless of the foregoing detailed description of exemplary embodiments of the invention, the optimum dimensional relationships for the individual components of the invention, including variations in size, shape, thickness, form, materials, function and manner of operation, assembly and use, as well as equivalents thereof, are deemed to be readily apparent and understood by those skilled in the art. Accordingly, equivalent relationships to those shown in the accompanying drawing figures and described in the written description are intended to be encompassed by the invention, the foregoing being considered as illustrative only of the general concept and principles of the invention. Furthermore, since numerous modifications and changes will readily occur to those skilled in the art, the exemplary embodiments disclosed herein are not intended to limit the invention to the specific configuration, construction, materials and operation shown and described. Instead, all reasonably predictable and suitable equivalents and obvious modifications to the invention should be construed as falling within the scope of the invention as defined by the appended claims given their broadest reasonable interpretation in view of the accompanying written description and drawings.

Claims

1. A rearview mirror with integrated sign, comprising: a housing; a mirror attached to the housing; a holder assembly attached to the housing, wherein the holder assembly includes a sign shaft, and wherein the sign shaft is capable of rotational movement; a sign attached to the sign shaft, wherein the sign includes indicia; a slot in the housing adapted for allowing the sign to move there-through; and at least one drive assembly functionally attached to the sign shaft and capable of rotating the sign shaft in a first direction to roll up the sign on the sign shaft in a stowed position and rotating the sign shaft in a second direction to deploy the sign in a deployed position whereby indicia on the sign are external of the housing; and at least one clamp attached to the housing, each of the at least one clamp being adapted for attachment to a rearview mirror, wherein each of the at least one clamp has a moveable jaw and a fixed jaw.
2. The rearview mirror with integrated sign of claim 1, wherein the moveable jaw is capable of being moved towards and away from the stationary jaw.
3. The rearview mirror with integrated sign of claim 2, further comprising a spring wherein the spring is attached to the moveable jaw and biases the moveable jaw towards the stationary jaw.
4. The rearview mirror with integrated sign of claim 3, wherein the spring is attached to the housing.
5. The rearview mirror with integrated sign of claim 2, further comprising a spring have a first end section, a second end portion and an intermediate section; wherein the at least one clamp is a first clamp and a second clamp; wherein the intermediate section of the spring is attached to the housing; wherein the first end section of the spring is attached to the moveable jaw of the first clamp and biases the moveable jaw towards the stationary jaw of the first clamp; and wherein the second end section of the spring is attached to the second clamp and the biases the moveable jaw towards the stationary jaw of the second clamp.
6. A rearview mirror with integrated sign, comprising: a housing; a mirror attached to the housing; a holder assembly attached to the housing, wherein the holder assembly includes a sign shaft, and wherein the sign shaft is capable of rotational movement; a sign attached to the sign shaft, wherein the sign includes indicia; a slot in the housing adapted for allowing the sign to move there-through; at least one drive assembly functionally attached to the sign shaft and capable of rotating the sign shaft in a first direction to roll up the sign on the sign shaft in a stowed position and rotating the sign shaft in a second direction to deploy the sign in a deployed position whereby indicia on the sign are external of the housing; and wherein the at least one drive assembly is a power drive assembly comprising a motor for rotating the sign shaft and a switch for selectively turning the motor on or off, and for selectively rotating the sign shaft in the first direction or the second direction.
7. The rearview mirror with integrated sign of claim 6, further including at least one sensor, wherein the at least one sensor signals to stop the motor when the sign reaches the deployed position when deploying the sign, and signals to stop the motor when the sign reaches the stowed position when retracting the sign.
8. The rearview mirror with integrated sign of claim 7, wherein each of the at least one sensor is an inferred sensor.
9. The rearview mirror with integrated sign of claim 6, wherein the at least one sensor is a first inferred sensor and a second inferred sensor; wherein the sign includes a first opening in a lower portion of the sign and a second opening in an upper portion of the sign; wherein upon activation of the motor to rotate the sign shaft in the first direction, the sign shaft will continue to rotate in the first direction and roll up the sign until the sensor detects the first opening and signals the motor to stop; and wherein upon activation of the motor to rotate the sign shaft in the second direction, the sign shaft will continue to rotate in the second direction and deploy the sign until the sensor detects second opening and signals the motor to stop.
10. A rearview mirror with integrated sign, comprising: a housing; a mirror attached to the housing;

a holder assembly attached to the housing, wherein the holder assembly includes a sign shaft, and wherein the sign shaft is capable of rotational movement; a sign attached to the sign shaft, wherein the sign includes indicia; a slot in the housing adapted for allowing the sign to move there-through; at least one drive assembly functionally attached to the sign shaft and capable of rotating the sign shaft in a first direction to roll up the sign on the sign shaft in a stowed position and rotating the sign shaft in a second direction to deploy the sign in a deployed position whereby indicia on the sign are external of the housing; and wherein the drive assembly is a manual drive assembly comprising a manually rotated scroll wheel for selectively rotating the sign shaft in the first direction or the second direction.

11. The rearview mirror with integrated sign of claim 10, wherein the scroll wheel includes a gear which engages a gear on the sign shaft whereby rotation of the scroll wheel in a first direction imparts rotation of the sign shaft in the first direction and rotation of the scroll wheel in a second direction imparts rotation of the sign shaft in the second direction.

12. The rearview mirror with integrated sign of claim 11, wherein the housing includes a slot; and wherein a portion of the scroll wheel is within the housing and a portion of the scroll wheel extends through the slot for being accessible by a person for manually operating the scroll wheel.

13. A rearview mirror with integrated sign, comprising: a housing; a mirror attached to the housing; a holder assembly attached to the housing, wherein the holder assembly includes a sign shaft, and wherein the sign shaft is capable of rotational movement; a sign attached to the sign shaft, wherein the sign includes indicia; a slot in the housing adapted for allowing the sign to move there-through; at least one drive assembly functionally attached to the sign shaft and capable of rotating the sign shaft in a first direction to roll up the sign on the sign shaft in a stowed position and rotating the sign shaft in a second direction to deploy the sign in a deployed position whereby indicia on the sign are external of the housing; and wherein the at least one drive assembly is a power drive assembly and a manual drive assembly; wherein the power drive assembly includes a motor for rotating the sign shaft and a switch for selectively turning the motor on or off, and for selectively rotating the sign shaft in the first direction or the second direction; and wherein the manual drive assembly includes a manually rotated scroll wheel for selectively rotating the sign shaft in the first direction or the second direction.

14. The rearview mirror with integrated sign of claim 13, further comprising at least one clamp attached to the housing, each of the at least one clamp being adapted for attachment to a rearview mirror.

15. The rearview mirror with integrated sign of claim 14, wherein the power drive assembly further includes at least one sensor, wherein the sensor signals to stop the motor when the sign reaches the deployed position when deploying the sign, and signals to stop the motor when the sign reaches the stowed position when retracting the sign.

16. The rearview mirror with integrated sign of claim 15, wherein the at least one sensor is a first inferred sensor and a second inferred sensor; wherein the sign includes a first opening in a lower portion of the sign and a second opening in an upper portion of the sign; wherein upon activation of the motor to rotate the sign shaft in the first direction, the sign shaft will continue to rotate in the first direction and roll up the sign until the sensor detects the first opening and signals the motor to stop; and wherein upon activation of the motor to rotate the sign shaft in the second direction, the sign shaft will continue to rotate in the second direction and deploy the sign until the sensor detects second opening and signals the motor to stop.

17. The rearview mirror with integrated sign of claim 16, wherein the scroll wheel includes a gear which engages a gear on the sign shaft whereby rotation of the scroll wheel in a first direction imparts rotation of the sign shaft in the first direction and rotation of the scroll wheel in a second direction imparts rotation of the sign shaft in the second direction.

18. The rearview mirror with integrated sign of claim 17, wherein the housing includes a slot; and

wherein a portion of the scroll wheel is within the housing and a portion of the scroll wheel extends through the slot for being accessible by a person for operating the scroll wheel.
